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The World on a String

THE FABRIC ACCORDING TO STRING THEORY

Imagine a universe in which to understand anything you'd need to understand everything. A universe in which to say anything about why a planet orbits a star, about why a baseball flies along a particular trajectory, about how a magnet or a battery works, about how light and gravity operate—a universe in which to say anything about anything—you would need to uncover the most fundamental laws and determine how they act on the finest constituents of matter. Thankfully, this universe is not our universe.

If it were, it's hard to see how science would have made any progress at all. Over the centuries, the reason we've been able to make headway is that we've been able to work piecemeal; we've been able to unravel mysteries step by step, with each new discovery going a bit deeper than the previous. Newton didn't need to know about atoms to make great strides in understanding motion and gravity. Maxwell didn't need to know about electrons and other charged particles to develop a powerful theory of electromagnetism. Einstein didn't need to address the primordial incarnation of space and time to formulate a theory of how they curve in the service of the gravitational force. Instead, each of these discoveries, as well as the many others that underlie our current conception of the cosmos, proceeded within a limited context that unabashedly left many basic questions unanswered. Each discovery was able to contribute its own piece to the puzzle, even though no one knew—and we still don't know—what grand synthesizing picture comprises all the puzzle's pieces.

A closely related observation is that although today's science differs sharply from that of even fifty years ago, it would be simplistic to summarize scientific progress in terms of new theories overthrowing